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Breaking Barriers: Addressing Algorithmic Bias For Equitable Cancer Care Access

PREVENT *– Predicting Reactions and Enhancing Vigilance in Elderly Insulin Treatment*

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Introduction

By 2050 Ireland's population aged 65 and over is estimated to be over 1.5 million [1]. With age, the prevalence of adverse drug reactions (ADRs) rises significantly. A study by Cork researchers revealed that one in four older adults experienced at least one ADR over six years. While most ADRs were mild, 11% were of moderate severity, and some cases led to emergency hospital admissions. Alarmingly, older adults are twice as likely to be hospitalized due to ADRs compared to younger individuals [2].

This project aims to address this growing concern by developing a predictive model to evaluate the safety of approved drugs for older adults using insulin. By leveraging predictive modeling, we strive to enhance patient safety and reduce the risk of adverse drug reaction.

Research Objectives

Our study focuses on advancing methods to predict and mitigate adverse drug reactions to enhance patient safety. The key objectives of our research are:

- **Develop an Interface**: Create an accessible and user-friendly interface to assist healthcare providers and individuals in predicting possible adverse reactions based on patient profiles.
- **Predictive Model Exploration**: Investigate the effectiveness of utilizing chained predictive models to assess the accuracy and reliability of reaction predictions.
- **Performance Assessment**: Evaluate the performance of our predictive models to ensure their accuracy and robustness in identifying adverse reactions.

Methodology

1. Extracted data from the F.A.E.R.S database [3] for patients aged 65+ using a type of insulin.
2. Designed a neural network model to classify reactions as serious or non-serious based on patient profiles.
3. Built a multi-class classifier to predict specific adverse reactions, integrating the output from the first model and patient profiles.

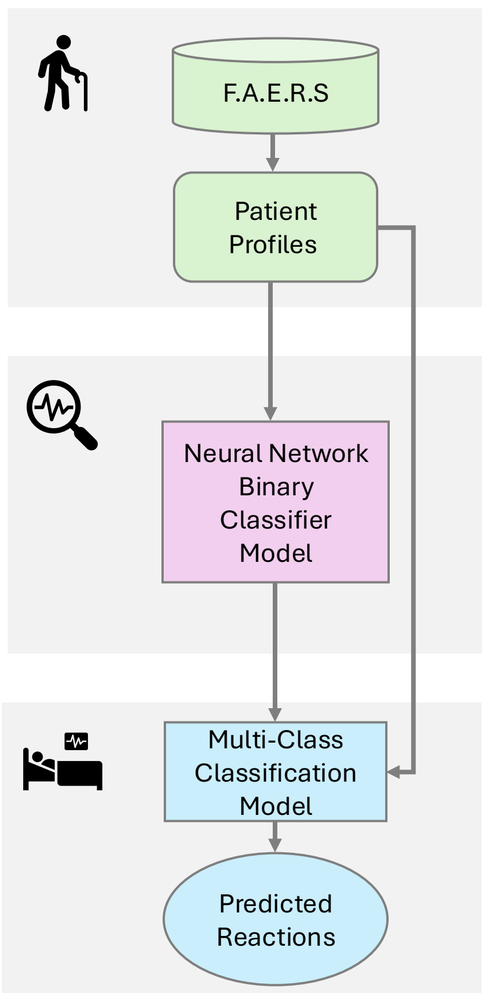

Fig.1 Displays the methodology for the project.

Results

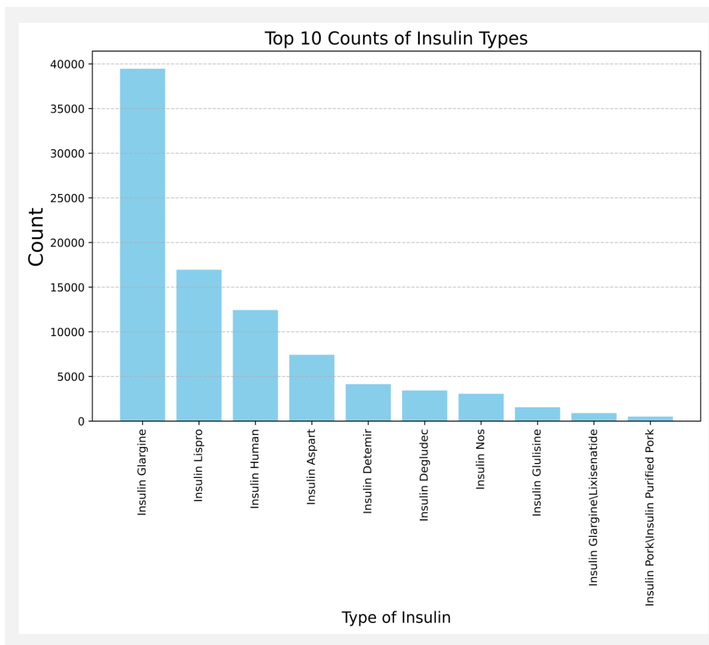

Fig.2. Top 10 count of reactions per insulin type

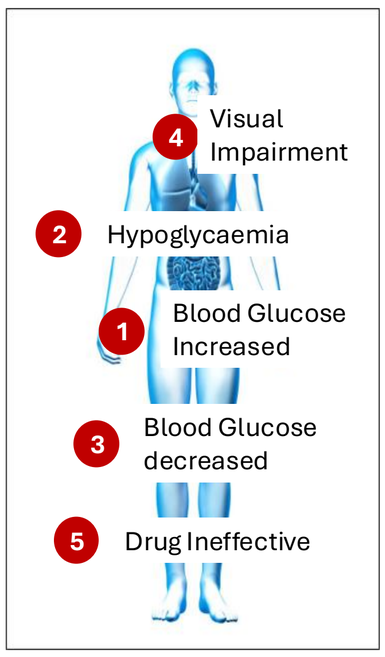

Fig.3. Example of UI with top 5 reactions

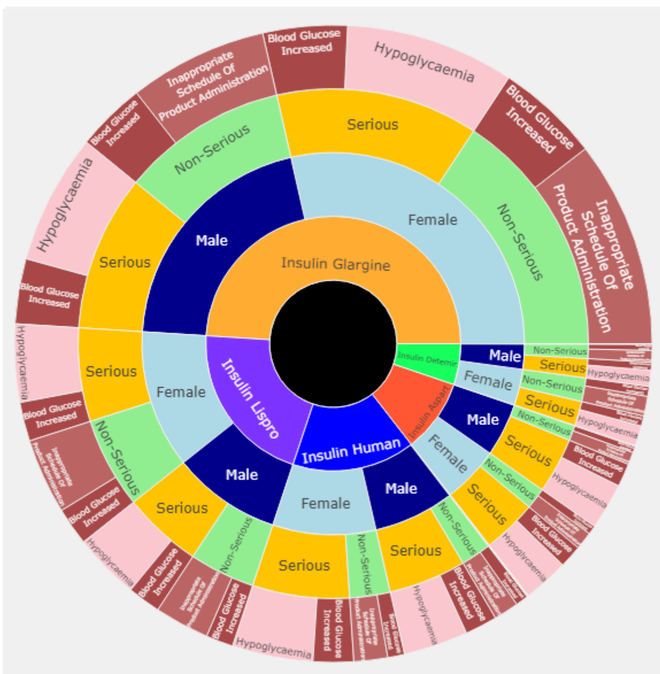

Fig.4 Two most common reactions per severity by sex, for top 5 most used insulin types

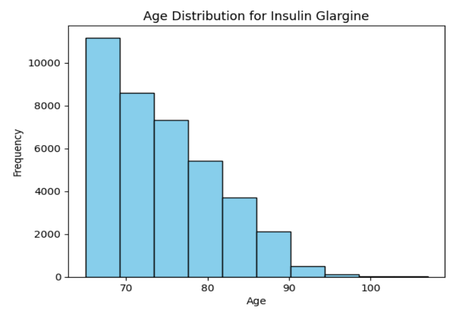

Fig.5. Age distribution for Insulin type Glargine

Fig.6 Age distribution for Insulin type Lispro

Conclusions

Our research highlights the potential of predictive modeling to enhance patient safety by identifying and mitigating adverse drug reactions. The development of a user-friendly interface demonstrates practical applications for healthcare providers and patients.

Future Work:

- Expand datasets to improve generalizability across diverse populations.
- Enhance predictive accuracy using advanced modeling techniques.
- Address biases and ethical considerations in handling sensitive patient data.

References/Acknowledgements

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[2]Doherty, A.S. et al. (2022) "Adverse drug reactions and associated patient characteristics in older community-dwelling adults: a 6-year prospective cohort study," British Journal of General Practice. Royal College of General Practitioners. Available at: https://doi.org/10.3399/bjgp.2022.0181.

[3]U.S. Food and Drug Administration (2025) FDA Adverse Event Reporting System (FAERS). Available at:[https://fis.fda.gov/sense/app/95239e26-e0be-42d9-a960-9a5f7f1c25ee/sheet/6b5a13 5f-f451-45be-893d-20aaee34e28e/state/analysis] (Accessed: 25 March 2025).

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